Starting at $t = 0$:

$$v_0^\pi \;=\; \mathrm{E}_\pi \left[ \sum_{t=0}^{\infty} \gamma^t \; R_{t+1} \right] \;=\; \mathrm{E}_\pi \left[ G_0 \right] \;, \tag{A7}$$

the optimal state-value function $v_*$ and policy $\pi_*$ are

$$v_*(s) \;=\; \max_\pi v^\pi(s) \;, \tag{A8}$$

$$\pi_* \;=\; \arg\max_\pi v^\pi(s) \;\text{ for all } s \;. \tag{A9}$$

The action-value function $q^\pi(s,a)$ for policy $\pi$ is the expected return when starting from $S_t = s$, taking action $A_t = a$, and following policy $\pi$:

$$q^\pi(s,a) \;=\; \mathrm{E}_\pi \left[ G_t \mid S_t = s, A_t = a \right] \;=\; \mathrm{E}_\pi \left[ \sum_{k=0}^{\infty} \gamma^k \; R_{t+k+1} \mid S_t = s, A_t = a \right] \;. \tag{A10}$$

The optimal action-value function $q_*$ and policy $\pi_*$ are

$$q_*(s,a) \;=\; \max_\pi q^\pi(s,a) \;, \tag{A11}$$

$$\pi_* \;=\; \arg\max_\pi q^\pi(s,a) \;\text{ for all } (s,a) \;. \tag{A12}$$

The optimal action-value function $q_*$ can be expressed via the optimal value function $v_*$:

$$q_*(s,a) \;=\; \mathrm{E} \left[ R_{t+1} \;+\; \gamma \, v_*(S_{t+1}) \mid S_t = s, A_t = a \right] \;. \tag{A13}$$

The optimal state-value function $v_*$ can be expressed via the optimal action-value function $q_*$ using the optimal policy $\pi_*$:

$$\begin{aligned} v_*(s) \;=\; & \max_a q^{\pi_*}(s,a) \;=\; \max_a \mathrm{E}_{\pi_*} \left[ G_t \mid S_t = s, A_t = a \right] \;= \\ & \max_a \mathrm{E}_{\pi_*} \left[ R_{t+1} \;+\; \gamma \, G_{t+1} \mid S_t = s, A_t = a \right] \;= \\ & \max_a \mathrm{E} \left[ R_{t+1} \;+\; \gamma \, v_*(S_{t+1}) \mid S_t = s, A_t = a \right] \;. \end{aligned} \tag{A14}$$

**Finite time horizon and no discount.** We consider a **finite** time horizon, that is, we consider only episodes of length $T$, but may receive reward $R_{T+1}$ at episode end at time $T+1$. The finite time horizon MDP creates a sequence

$$(S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \ldots, S_{T-1}, A_{T-1}, R_T, S_T, A_T, R_{T+1}) \;. \tag{A15}$$

Furthermore, we do not discount future rewards, that is, we set $\gamma = 1$. The return $G_t$ from time $t$ to $T$ is the sum of rewards:

$$G_t \;=\; \sum_{k=0}^{T-t} R_{t+k+1} \;. \tag{A16}$$

The state-value function $v$ for policy $\pi$ is

$$v^\pi(s) \;=\; \mathrm{E}_\pi \left[ G_t \mid S_t = s \right] \;=\; \mathrm{E}_\pi \left[ \sum_{k=0}^{T-t} R_{t+k+1} \mid S_t = s \right] \tag{A17}$$

and the action-value function $q$ for policy $\pi$ is

$$\begin{aligned} q^\pi(s,a) \;=\; & \mathrm{E}_\pi \left[ G_t \mid S_t = s, A_t = a \right] \;=\; \mathrm{E}_\pi \left[ \sum_{k=0}^{T-t} R_{t+k+1} \mid S_t = s, A_t = a \right] \\ =\; & \mathrm{E}_\pi \left[ R_{t+1} \;+\; G_{t+1} \mid S_t = s, A_t = a \right] \\ =\; & \sum_{s',r} p(s', r \mid s, a) \left[ r \;+\; \sum_{a'} \pi(a' \mid s') \, q^\pi(s', a') \right] \;. \end{aligned} \tag{A18}$$

From the Bellman equation Eq. (A18), we obtain:

$$\sum_{s'} p(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\, q^\pi(s',a') \;=\; q^\pi(s,a) \;-\; \sum_r r\, p(r \mid s,a)\;, \tag{A19}$$

$$\mathrm{E}_{s',a'}\left[q^\pi(s',a') \mid s,a\right] \;=\; q^\pi(s,a) \;-\; r(s,a)\;. \tag{A20}$$

The expected return at time $t=0$ for policy $\pi$ is

$$v_0^\pi \;=\; \mathrm{E}_\pi\left[G_0\right] \;=\; \mathrm{E}_\pi\left[\sum_{t=0}^{T} R_{t+1}\right]\;, \tag{A21}$$

$$\pi^* \;=\; \operatorname*{argmax}_\pi\; v_0^\pi\;.$$

The agent may start in a particular starting state $S_0$ which is a random variable. Often $S_0$ has only one value $s_0$.

**Learning.** The **goal** of learning is to find the policy $\pi^*$ that maximizes the expected future discounted reward (the return) if starting at $t=0$. Thus, the optimal policy $\pi^*$ is

$$\pi^* \;=\; \operatorname*{argmax}_\pi\; v_0^\pi\;. \tag{A22}$$

We consider two learning approaches for $Q$-values: Monte Carlo and temporal difference.

**Monte Carlo (MC).** To estimate $q^\pi(s,a)$, MC computes the arithmetic mean of all observed returns $(G_t \mid S_t = s, A_t = a)$ in the data. When using Monte Carlo for learning a policy we use an exponentially weighted arithmetic mean since the policy steadily changes.
For the $i$th update Monte Carlo tries to minimize $\frac{1}{2}M(s_t,a_t)^2$ with the residual $M(s_t,a_t)$

$$M(s_t,a_t) \;=\; (q^\pi)^i(s_t,a_t) \;-\; \sum_{\tau=0}^{T-t-1} \gamma^\tau r_{t+1+\tau}\;, \tag{A23}$$

such that the update of the action-value $q$ at state-action $(s_t,a_t)$ is

$$(q^\pi)^{i+1}(s_t,a_t) \;=\; (q^\pi)^i(s_t,a_t) \;-\; \alpha\, M(s_t,a_t)\;. \tag{A24}$$

This update is called *constant-$\alpha$ MC* [128].

**Temporal difference (TD) methods.** TD updates are based on the Bellman equation. If $r(s,a)$ and $\mathrm{E}_{s',a'}\left[\hat{q}^\pi(s',a') \mid s,a\right]$ have been estimated, the $Q$-values can be updated according to the Bellman equation:

$$(\hat{q}^\pi)^{\mathrm{new}}(s,a) \;=\; r(s,a) \;+\; \gamma\, \mathrm{E}_{s',a'}\left[\hat{q}^\pi(s',a') \mid s,a\right]\;. \tag{A25}$$

The update is applying the Bellman operator with estimates $\mathrm{E}_{s',a'}\left[\hat{q}^\pi(s',a') \mid s,a\right]$ and $r(s,a)$ to $\hat{q}^\pi$ to obtain $(\hat{q}^\pi)^{\mathrm{new}}$. The new estimate $(\hat{q}^\pi)^{\mathrm{new}}$ is closer to the fixed point $q^\pi$ of the Bellman operator, since the Bellman operator is a contraction (see Section A7.1.3 and Section A7.1.2).
Since the estimates $\mathrm{E}_{s',a'}\left[\hat{q}^\pi(s',a') \mid s,a\right]$ and $r(s,a)$ are not known, TD methods try to minimize $\frac{1}{2}B(s,a)^2$ with the Bellman residual $B(s,a)$:

$$B(s,a) \;=\; \hat{q}^\pi(s,a) \;-\; r(s,a) \;-\; \gamma\, \mathrm{E}_{s',a'}\left[\hat{q}^\pi(s',a')\right]\;. \tag{A26}$$

TD methods use an estimate $\hat{B}(s,a)$ of $B(s,a)$ and a learning rate $\alpha$ to make an update

$$\hat{q}^\pi(s,a)^{\mathrm{new}} \;\leftarrow\; \hat{q}^\pi(s,a) \;-\; \alpha\, \hat{B}(s,a)\;. \tag{A27}$$

For all TD methods $r(s,a)$ is estimated by $R_{t+1}$ and $s'$ by $S_{t+1}$, while $\hat{q}^\pi(s',a')$ does not change with the current sample, that is, it is fixed for the estimate. However, the sample determines which $(s',a')$ is chosen. The TD methods differ in how they select $a'$. **SARSA** [105] selects $a'$ by sampling from the policy:

$$\mathrm{E}_{s',a'}\left[\hat{q}^\pi(s',a')\right] \;\approx\; \hat{q}^\pi(S_{t+1},A_{t+1})$$

and **expected SARSA** [63] averages over selections

$$\mathrm{E}_{s',a'}\left[\hat{q}^\pi(s',a')\right] \;\approx\; \sum_a \pi(a \mid S_{t+1})\, \hat{q}^\pi(S_{t+1},a).$$

It is possible to estimate $r(s,a)$ separately via an unbiased minimal variance estimator like the arithmetic mean and then perform TD updates with the Bellman error using the estimated $r(s,a)$ [103]. ***Q*-learning** [140] is an off-policy TD algorithm which is proved to converge [141, 20]. The proofs were later generalized [61, 133]. $Q$-learning uses

$$\mathrm{E}_{s',a'}\left[\hat{q}^{\pi}(s',a')\right] \;\approx\; \max_{a} \hat{q}(S_{t+1},a)\;. \tag{A28}$$

The action-value function $q$, which is learned by $Q$-learning, approximates $q_*$ independently of the policy that is followed. More precisely, with $Q$-learning $q$ converges with probability 1 to the optimal $q_*$. However, the policy still determines which state-action pairs are encountered during learning. The convergence only requires that all action-state pairs are visited and updated infinitely often.

## A2 Reward Redistribution, Return-Equivalent SDPs, Novel Learning Algorithms, and Return Decomposition

### A2.1 State Enriched MDPs

For MDPs with a delayed reward the states have to code the reward. However, for an immediate reward the states can be made more compact by removing the reward information. For example, states with memory of a delayed reward can be mapped to states without memory. Therefore, in order to compare MDPs, we introduce the concept of homomorphic MDPs. We first need to define a partition of a set induced by a function. Let $B$ be a partition of a set $X$. For any $x \in X$, we denote $[x]_B$ the block of $B$ to which $x$ belongs. Any function $f$ from a set $X$ to a set $Y$ induces a partition (or equivalence relation) on $X$, with $[x]_f = [x']_f$ if and only if $f(x) = f(x')$. We now can define homomorphic MDPs.

**Definition A1** (Ravindran and Barto [98, 99])**.** *An MDP homomorphism $h$ from an MDP $\mathcal{P} = (\mathcal{S}, \mathcal{A}, \mathcal{R}, p, \gamma)$ to an MDP $\tilde{\mathcal{P}} = (\tilde{\mathcal{S}}, \tilde{\mathcal{A}}, \tilde{\mathcal{R}}, \tilde{p}, \tilde{\gamma})$ is a a tuple of surjections $(f, g_1, g_2, \ldots, g_n)$ ($n$ is number of states), with $h(s,a) = (f(s), g_s(a))$, where $f : \mathcal{S} \to \tilde{\mathcal{S}}$ and $g_s : \mathcal{A}_s \to \tilde{\mathcal{A}}_{f(s)}$ for $s \in \mathcal{S}$ ($\mathcal{A}_s$ are the admissible actions in state $s$ and $\tilde{\mathcal{A}}_{f(s)}$ are the admissible actions in state $\tilde{s}$). Furthermore, for all $s, s' \in \mathcal{S}, a \in \mathcal{A}_s$:*

$$\tilde{p}(f(s') \mid f(s), g_s(a)) \;=\; \sum_{s'' \in [s']_f} p(s'' \mid s, a)\;, \tag{A29}$$

$$\tilde{p}(\tilde{r} \mid f(s), g_s(a)) \;=\; p(r \mid s, a)\;. \tag{A30}$$

*We use $[s]_f = [s']_f$ if and only if $f(s) = f(s')$.*

We call $\tilde{\mathcal{P}}$ the *homomorphic image* of $\mathcal{P}$ under $h$. For homomorphic images the optimal $Q$-values and the optimal policies are the same.

**Lemma A1** (Ravindran and Barto [98])**.** *If $\tilde{\mathcal{P}}$ is a homomorphic image of $\mathcal{P}$, then the optimal $Q$-values are the same and a policy that is optimal in $\tilde{\mathcal{P}}$ can be transformed to an optimal policy in $\mathcal{P}$ by normalizing the number of actions $a$ that are mapped to the same action $\tilde{a}$.*

Consequently, the original MDP can be solved by solving a homomorphic image.
Similar results have been obtained by Givan et al. using stochastically bisimilar MDPs: "Any stochastic bisimulation used for aggregation preserves the optimal value and action sequence properties as well as the optimal policies of the model" [34]. Theorem 7 and Corollary 9.1 in Givan et al. show the facts of Lemma A1. Li et al. give an overview over state abstraction and state aggregation for Markov decision processes, which covers homomorphic MDPs [73].
A Markov decision process $\tilde{\mathcal{P}}$ is state-enriched compared to an MDP $\mathcal{P}$ if $\tilde{\mathcal{P}}$ has the same states, actions, transition probabilities, and reward probabilities as $\mathcal{P}$ but with additional information in its states. We define state-enrichment as follows:

**Definition A2.** *A Markov decision process $\tilde{\mathcal{P}}$ is* state-enriched *compared to a Markov decision process $\mathcal{P}$ if $\mathcal{P}$ is a homomorphic image of $\tilde{\mathcal{P}}$, where $g_{\tilde{s}}$ is the identity and $f(\tilde{s}) = s$ is not bijective.*

Being not bijective means that there exist $\tilde{s}'$ and $\tilde{s}''$ with $f(\tilde{s}') = f(\tilde{s}'')$, that is, $\tilde{\mathcal{S}}$ has more elements than $\mathcal{S}$. In particular, state-enrichment does not change the optimal policies nor the $Q$-values in the sense of Lemma A1.

**Proposition A1.** *If an MDP $\tilde{\mathcal{P}}$ is* state-enriched *compared to an MDP $\mathcal{P}$, then both MDPs have the same optimal $Q$-values and the same optimal policies.*